

TrES-2b

R-1039

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_160429 · created 2026-07-30T07:40:09+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457507.90766 +/- 0.00041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1483 +/- 0.0073
Transit depth (Rp/Rs)^2	2.2 +/- 0.22 [%]
Orbital Inclination (inc)	84.0 +/- 0.29
Airmass coefficient 1 (a1)	1.315 +/- 0.011
Airmass coefficient 2 (a2)	-0.2073 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	None
Optimal Aperture	4.04
Optimal Annulus	4.71
Transit Duration (day)	0.0798 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1483 ± 0.0073 vs 0.1254 (deviation 2.93σ)
Duration fit vs published	0.0798 d vs 0.0746 d (deviation 1.16σ)
Mid-time O–C	1.1 min
Depth SNR	19.0
Residual scatter	0.82 %

Light curve

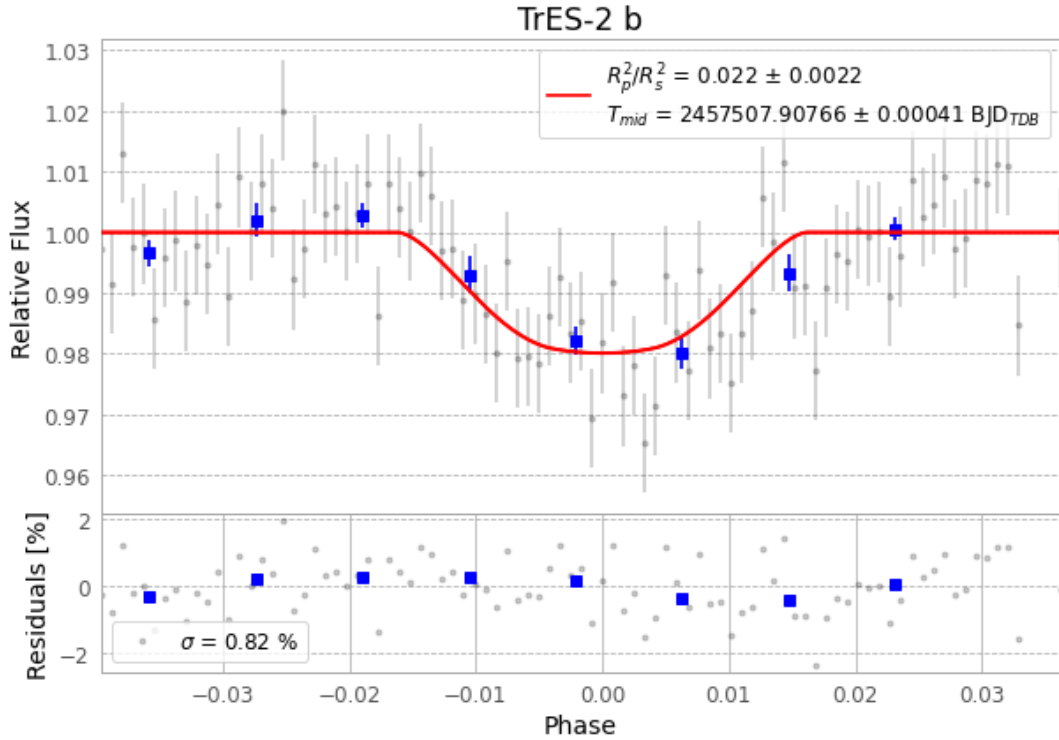


Figure 1 — FinalLightCurve_TrES-2 b_2016-04-29.png (SHA-256 002f74d1efbec533...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 162], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 73082d13b1183d7a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 77e40fd

Data provenance

90 FITS frames (59 MB), TRES-2160429072412.FITS → TRES-2160429120011.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-160429.log (39eeebd72bd9...), FinalLightCurve_TrES-2 b_2016-04-29.png (002f74d1efbe...), FinalLightCurve_TrES-2 b_2016-04-29.csv (38eb6e8ad5e3...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 07:40Z.